```
  GNU nano 7.2
#include <stdio.h>
#include <string.h>

#define MAX 5
#define SIZE 100

int main() {
    char history[MAX][SIZE];
    char command[SIZE];
    int count = 0, i;

    while (1) {
        printf("Enter command (exit to stop): ");
        if (!fgets(command, SIZE, stdin))
            break;

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
            break;

        if (strlen(command) == 0)
            continue;

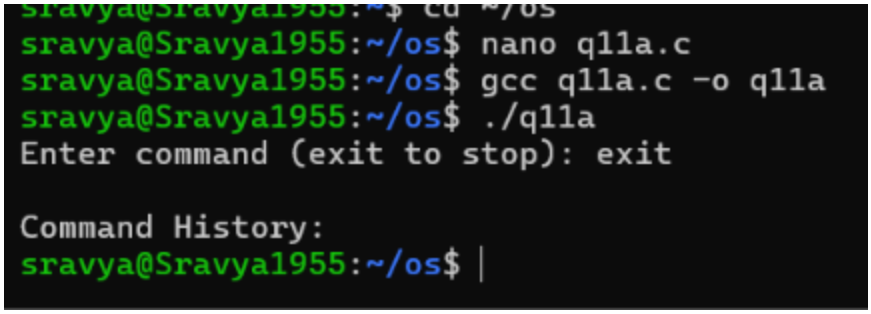
        strcpy(history[count % MAX], command);
        count++;
    }

    printf("\nCommand History:\n");

    int start = count > MAX ? count - MAX : 0;

    for (i = start; i < count; i++)
        printf("%d. %s\n", i + 1, history[i % MAX]);

    return 0;
}
```


sravya@Sravya1955:~$ cd ~/os
sravya@Sravya1955:~/os$ nano q11a.c
sravya@Sravya1955:~/os$ gcc q11a.c -o q11a
sravya@Sravya1955:~/os$ ./q11a
Enter command (exit to stop): exit
Command History:
sravya@Sravya1955:~/os$

```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    pipe(fd);

    pid_t pid = fork();

    if (pid == 0) {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", NULL);
        perror("execlp");
        _exit(1);
    } else if (pid > 0) {
        pid_t pid2 = fork();

        if (pid2 == 0) {
            dup2(fd[0], STDIN_FILENO);
            close(fd[0]);
            close(fd[1]);
            execlp("wc", "wc", "-l", NULL);
            perror("execlp");
            _exit(1);
        }

        close(fd[0]);
        close(fd[1]);
        waitpid(pid, NULL, 0);
        waitpid(pid2, NULL, 0);
    } else {
        perror("fork");
        return 1;
    }
```

```
sravya@Sravya1955:~/os$ nano q11b.c
sravya@Sravya1955:~/os$ gcc q11b.c -o q11b
sravya@Sravya1955:~/os$ ./q11b
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```